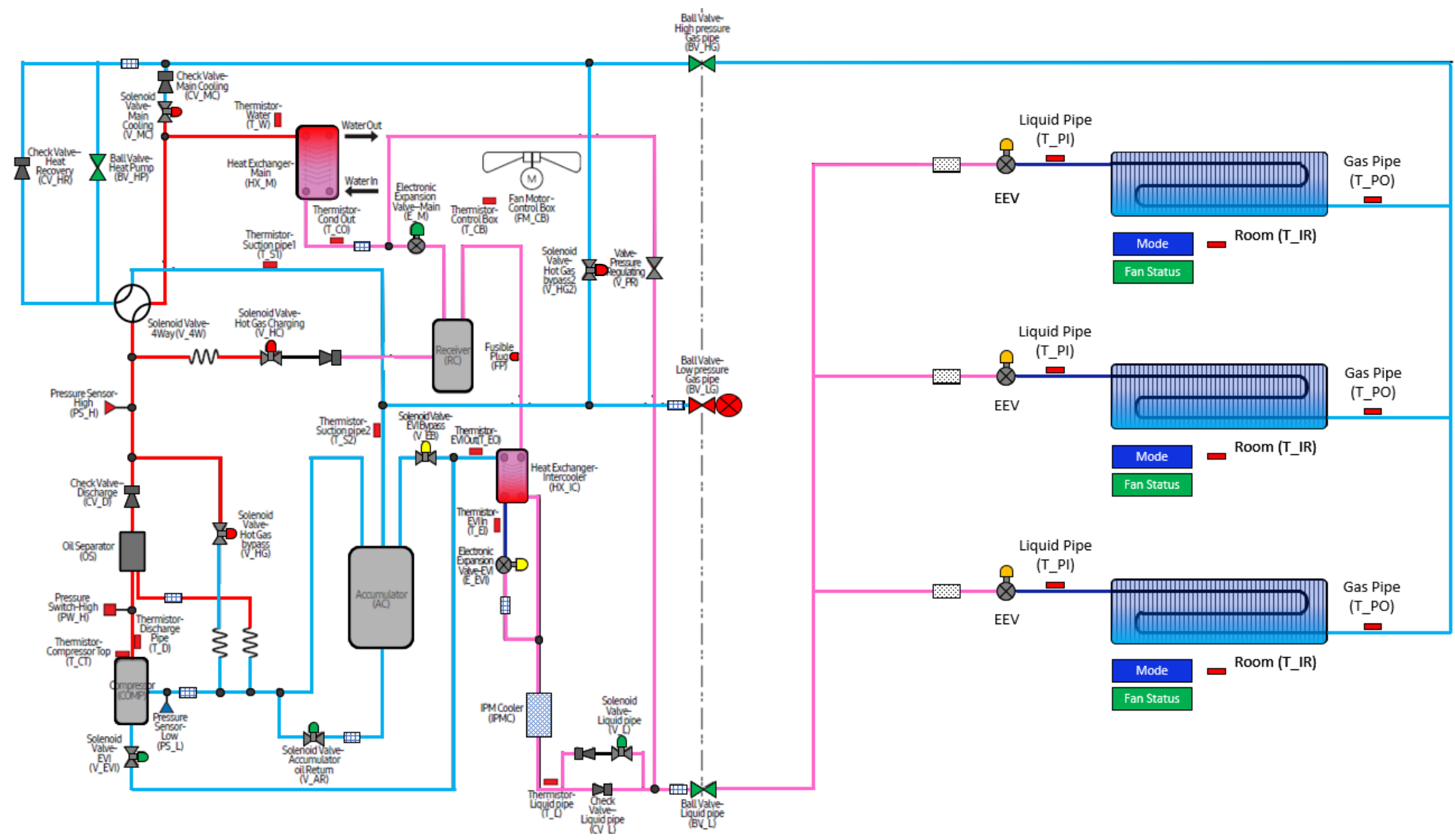


DVM S 3 Phase Water
Heat Pump Operation Evaluation
Cooling Mode



Outdoor Unit		
Component	Target	Reading
Water Out Temp.		
High pressure	427 Psig	
H.P Sat. Temp		
Discharge Temp.	150 ⁰ -212 ⁰	
IPM Temp,	Below 194 ⁰	
Low pressure		
L.P. Sat. Temp	OAT – (2 ⁰ -10 ⁰)	
Suction 1 Temp.		
Suction 2 Temp		
Super Heat.	3.6 ⁰ or 9 ⁰	
Min EEV Position		
Cond. Out Temp.	OAT – (2 ⁰ – 35 ⁰)	
Liquid Line Temp.		
Sub-Cooling		
Indoor Units		
Component	Target	Reading
IDU 1 EEV		
IDU 1 SH	3.6 (0-7)	
IDU 1 T PI		
IDU 1 T PO		
IDU 2 EEV		
IDU 2 SH	3.6 (0 - 7)	
IDU 2 T PI		
IDU 2 T PO		
IDU 3 EEV		
IDU 3 SH	3.6 (0-7)	
IDU 3 T PI		
IDU 3 T PO		

Comments:

DVM S 3 Phase Water
Heat Pump Target Operation

Water Source Outdoor Unit				
Component	Abbreviation		Cooling Mode	Comments
Compressor	Comp	Operation	Target low pressure control (default: 113.8psig)	
		Range	14~140Hz	
Main EEV	E_M	Operation	Full Open	
		Range	2000 Steps	
EVI EEV	E_EVI	Operation	*Subcooling control (default: SC=36°F)	
		Range	0-480step	
4 Way Valve	V_4W	Operation	Off	
EVI By-Pass Solenoid Valve	V_EB	Operation	On (Open): *T_W > 104°F or 50Hz↓	Gas goes to the Accumulator
			Off (Close): T_W < 100.4°F↓ and 75 Hz↑	Refrigerant goes to vapor injection circuit DSH control
EVI Solenoid Valve	V_EVI	Operation	On (Open)	
Hot Gas By-Pass	V_HG	Operation	Off (Closed)	
Accumulator Oil Return Solenoid Valve	V_AR	Operation	On (Open)	Off during Start Up, On 2 min. after compressor start
Hot Gas By-Pass 2 (HR Only)	V_HG2	Operation	Off Closed	Opens after 20 minutes of continuous run time in cooling mode
Liquid Line Solenoid Valve	V_L	Operation	Off (Closed)	
Main Cooling Solenoid	VMC	Operation	Off (Closed)	
Hot Gas Charging Solenoid Valve	V_HC	Operation	Off (Closed)	Open when High Pressure is low?
Mode Change Unit (MCU)				
Component	Abbreviation		Cooling Mode	Comments
Port Heating Solenoid	Heat	Operation	Off (Closed)	
Port Cooling Solenoid	Cool	Operation	On (Open)	
Port EEV	EEV	Operation	Open 480 Steps	Open when a zone calls for cooling
		Range	0 - 480 Steps	
Sub-cooling EEV	SC_EEV	Operation	Closed 0 Steps	Active only in Heating Main and Cooling Main
		Range	480 Steps	
Sub-Cooler In Temperature		Operation		
Sub-Cooler Out Temperature		Operation	Close to Suction Header Temperature	
Indoor Unit IDU				
Component	Abbreviation		Cooling Mode	Comments
Room Temperature	Room Temp.	Operation	67-80 Degrees	
Liquid Pipe Thermistor	EVA In 1	Operation	Evaporator Temperature	Used to calculate Evaporator Superheat
Gas Pipe Thermistor	EVA Out 1	Operation	Measures Suction Pipe Temperature	Used to calculate Evaporator Superheat
EEV	EEV 1	Operation	Modulates to maintain Superheat of 3.6 degrees F	Superheat = to Pipe out Temp - Pipe in Temp.
		Range	0 - 2000 Steps (0 Steps when zone is satisfied)	Typical range between 250 and 1500 Steps
IDU Fan	Fan Speed	Range	Low, Medium, High, Auto	
		Operation	Low Speed: Ts - Ti ≤ 1.8°F	TS - Set Temperature
			Medium Speed: 1.8°F < Ts - Ti ≤ 5.4°F	Ti Indoor Temperature
			High Speed 5.4°F < Ts - Ti	